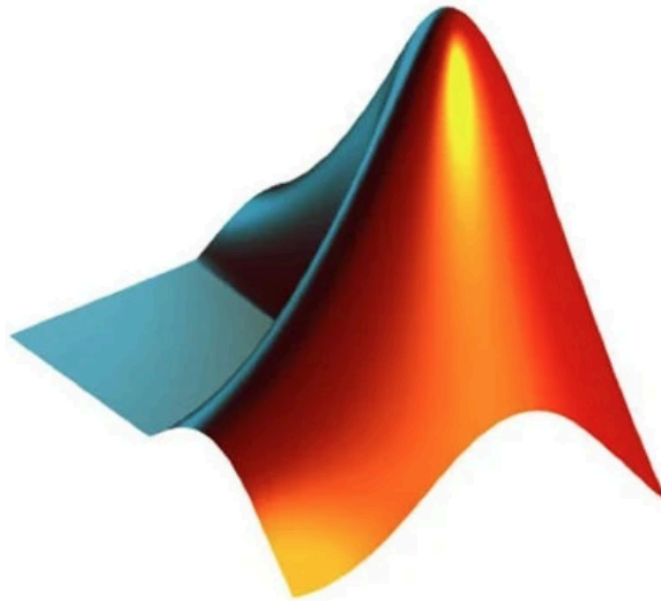


MATH 284 MATLAB Project #2 - Matrix Algebra Workbench



MATLAB®

Row Matrix $\begin{bmatrix} 3 & 5 & 7 \end{bmatrix}$	Column Matrix $\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$	Square Matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
Diagonal Matrix $\begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}$	Scalar Matrix $\begin{bmatrix} 5 & 0 \\ 5 & 5 \end{bmatrix}$	Identity Matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
Zero or Null Matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	Upper Triangular Matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$	Lower Triangular Matrix $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$
Symmetric Matrix $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$	Skew-Symmetric Matrix $\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$	Singular Matrix $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$
Non-Singular Matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	Orthogonal Matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	Equal Matrices $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$

Project Objective

Develop a MATLAB application that allows users to create, manipulate, analyze, and visualize matrices while exploring the major concepts of matrix algebra.

Instead of only computing solutions, the program explains matrix properties and whether a matrix satisfies important linear algebra conditions.

Concepts And Textbook Chapters/Sections Covered

Chapter 2 - Matrix Algebra

- 2.1 Matrix Operations
- 2.2 Inverse of a Matrix
- 2.3 Characterizations of Invertible Matrices
- 2.8 Subspaces of $\mathbb{R}^n$
- 2.9 Dimension and Rank

- Matrix operations
- Matrix inverse
- Invertible Matrix Theorem
- Subspaces of $\mathbb{R}^n$
- Dimension
- Rank

GUI Layout

Matrix Algebra Workbench	
Matrix A	Matrix B
[]	[]
Rows: []	Columns: []
Matrix Operations	
[+] [-] [Multiply] [Transpose]	
[Inverse] [Determinant] [Rank]	
Matrix Analysis	
Is Invertible?	
Rank	
Nullity	
Dimension	
Independent Columns?	
Visualization	
Column Space / Transformation / Basis Display	
Advanced Tools	
[Invertible Matrix Test]	
[Subspace Explorer]	
[Column Space]	
[Row Space]	
[Null Space]	

Necessary MATLAB Files For This Project

MatrixAlgebraWorkbenchApp.m

Responsibilities:

- *Build GUI*
- *Handle user interaction*
- *Display results*
- *Connect all analysis modules*

MatrixAnalyzer.m

Stores the following information about the matrix:

- *inverse*
- *determinant*
- *rank*
- *dimension*

InvertibilityChecker.m

Checks whether

- *determinant $\neq 0$*
- *inverse exists*
- *full rank*
- *pivot in every row*
- *pivot in every column*
- *columns are linearly independent*
- *unique solution for $Ax=b$*

SubspaceExplorer.m

Computes:

- *Column Space*
- *Row Space*
- *Null Space*

RankDimensionAnalyzer.m

Computes:

- *Rank*
- *Nullity*
- *Dimension*

This is used for the Rank-Nullity Theorem.

MatrixVisualizer.m

Responsible for visualization.

Necessary MATLAB Methods For This Project

createComponents()

performOperation()

analyzeMatrix()

displayResults()

clearWorkspace()

add()

subtract()

multiply()

transpose()

inverse()

calculateRank()

calculateDeterminant()

isInvertible()

findColumnSpace()

findRowSpace()

findNullSpace()

plotBasis()

plotColumnVectors()

plotTransformation()

plotSubspace()

GUI Functions

- Add, subtract, multiply, transpose
- Compute inverse and verify $AA^{-1} = I$
- Interactive invertibility checker
- Column, row, and null space explorer
- Rank and dimension calculator

Pseudocode

START PROGRAM

Create GUI

Initialize:

Matrix A

Matrix B

Results panel

Plot panel

WHEN user selects Matrix Operation

Read matrices

Determine operation

Perform calculation

Display result

WHEN Analyze Matrix selected

Compute determinant

Compute inverse

Compute rank

Check invertibility

Display all properties

WHEN Subspace Explorer selected

Compute:

Column Space

Row Space

Null Space

Display basis

Plot subspace (if possible)

WHEN Rank Tool selected

Compute rank

Compute nullity

Display Rank-Nullity relationship

WHEN Clear selected

Reset GUI

END PROGRAM

$$\begin{bmatrix} 0 & 1 & 8 & 9 \\ 1 & 9 & 7 & 3 \\ 9 & 8 & 2 & 5 \\ 1 & 7 & 3 & 8 \end{bmatrix}$$

Row A

Col A

Nul A

$$\begin{bmatrix} 5 & 8 & 1 & 0 \\ 2 & 3 & 5 & 0 \\ 9 & 8 & 0 & 0 \\ 6 & 7 & 2 & 4 \end{bmatrix}$$

linear algebra

$$\lambda_1 = 3 \quad \lambda_2 = 7$$

$$\det(\mathbf{A}) = 0$$

$$\begin{bmatrix} 5 & 8 & 1 & 0 \\ 2 & 3 & 5 & 0 \\ 9 & 8 & 0 & 0 \\ 6 & 7 & 2 & 4 \end{bmatrix}$$

[9 8 0 0]
[6 5 2 4]

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